

INTISHAR-UL ISLAM

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Summary

Senior Software Engineer (Manager) with 7+ years building production mobile, desktop, and backend systems in Flutter and Rust. Currently at **AKIJ iBOS** leading a 4–6 engineer team across four products: real-time endpoint monitoring on ~1,030 deployments, VoIP across AKIJ Group, and a public OTA travel app; rewrote a Rust gRPC flight-search engine from 20–30s to 4–12s. Independent portfolio spans multi-tenant restaurant and inventory SaaS platforms (Rust + Flutter) and complex PostgreSQL schema design (111+ Prisma models). Previously shipped GariBook (500K+ Play Store downloads) and led Probashi (10K downloads in 3–5 months) at Algorithm Generation Limited. CSE graduate, United International University; IEEE publication (RAAICON 2019).

Skills

Languages: Dart, Rust, Java, C / C++, Python, .NET

Mobile (Flutter): Riverpod, Provider, Bloc, GetX, Dart FFI to Rust / C++, Method Channels, Workmanager, custom painting / animation

Backend & Desktop (Rust): Axum, SQLx, Tonic gRPC, Tauri, libp2p, Wasm, rayon, async / multi-threading, Windows FFI

Realtime & Comm: WebSocket, gRPC streaming, SIP, WebRTC, RTMP

Databases: PostgreSQL, ClickHouse, MySQL, MongoDB, SQLite, Isar, Hive

Cloud & Tooling: Docker, AWS S3, Cloudinary, Stripe, SSL Commerz, Firebase, Git, LaTeX

Experience

AKIJ iBOS Limited

Oct 2024 – Present

Senior Software Engineer II (Manager)

- Lead a 4–6 engineer team owning four production products across the AKIJ Group: **Trackforce**, **iDialer**, **Akij Air**, and **Air Search**.
- **Trackforce** – Architected an endpoint activity-monitoring platform (Rust + Tauri desktop agent, Axum + ClickHouse backend) now deployed on ~1,030 endpoints inside AKIJ iBOS and commercialized for external sale; implemented system-level input/screen capture, real-time WebSocket ingestion, image processing pipeline (JPEG/PNG → WebP via rayon), and 16 schema migrations supporting columnar analytics.
- **Air Search** – **Reduced flight-search latency from 20–30s to 4–12s** (~60–80% faster) by rewriting the supplier-aggregation engine in Rust using Tonic gRPC bidirectional streaming, parallel HTTP fan-out across 6 supplier integrations, AWS S3 audit logging, and MySQL persistence.
- **iDialer (VoIP)** – Built the Flutter SIP companion app deployed to AKIJ Group calling agents; integrated SIP UA, native CallKit (iOS/Android), embedded OpenVPN client, Hive-backed offline auth state, conference calling, and Riverpod code-generated state management.
- **Akij Air** – Driving the legacy React Native → Flutter migration of the public OTA flight-booking app (currently in soft launch); enforcing strict clean architecture with Riverpod codegen, Freezed pattern matching, GoRouter auth guards, encrypted secure storage, and English/Bangla ARB i18n.
- **Daylens (Rust AI chat-agent / coworker – in active development)** – Building an internal AI assistant for AKIJ iBOS on top of the open-source Goose agent framework; customized for company workflow with per-user API keys for multiple AI providers (Anthropic, OpenAI, Google, etc.), local **Ollama** inference for training on internal company data, and MCP-based tool extensibility (shell execution, file editing, internal-service integrations).

Algorithm Generation Limited

Jun 2021 – Oct 2024

Senior Software Engineer (Sep 2023–Oct 2024) | Software Engineer (Sep 2022–Sep 2023) | Junior Software Engineer (Jun 2021–Sep 2022)

- **Probashi** – **Project Lead**. Shipped a Flutter marketplace + social platform connecting Bangladeshi expatriates with experts for paid consultations; 90+ API endpoints across 25 domains, real-time WebSocket chat with offline message queuing, multi-channel auth (email + phone OTP + Google/Facebook/LinkedIn OAuth), SSL Commerz payments, geolocation-based discovery, English/Bangla i18n. **Reached 10K downloads in the first 3–5 months post-launch**.
- **GariBook** – **500K+ Play Store downloads**. Co-built the ride-sharing rider and driver Flutter apps (246- and 303-file codebases) covering Google Maps real-time matching, WebSocket trip updates, Firebase messaging + analytics + remote config, image compression for driver documents, and in-app video.
- **AGGS** – Built a cross-platform live event scoring & streaming platform (mobile, tablet, web) – 786 Dart files across 14 feature modules with real-time WebSocket data flow, RTMP live broadcasting, embedded video playback, and responsive layouts with `go_router` web navigation.
- **File Share** – Architected a hybrid Flutter + Rust internal file-sync application. After profiling Dart-side bottlenecks in parallel filesystem traversal and native file watching, wrote a **2,200-line Rust core** (compiled to `cdylib`, linked into Flutter via Dart FFI) implementing CRC32 block-level change detection, debounced `notify` filesystem events, `rayon` parallel directory walks, multi-destination sync, and SQLite + FTS5 hash metadata.
- Additional shipped projects: **Clerk** (in-house finance ERP, 11 modules, Syncfusion charts, Hive offline DB, `workmanager` background sync), **POS** (deployed at 2 restaurants – barcode scanning, inventory, receipt printing), Meeting Schedule, AG E-commerce, Like Dislike (dating match app).

Food OS

Aug 2023 – Jan 2024 | Freelance, concurrent with AGL

Flutter Developer – collaboration with Abu Anwar (Google Developer Expert in Flutter, Bangladesh)

- Co-built a Flutter food-ordering and delivery app spanning auth (sign-in / sign-up / verification), restaurant browsing with categorized menus and item add-ons, cart with promo codes, Google Maps-integrated delivery location picker, multiple payment methods, loyalty points + redemption history, driver assignment, and live order-status tracking.

App Coder XYZ

Dec 2020 – May 2021

Junior Flutter Developer

- Contributed to **Teamwave**, a multi-module CRM / project-management platform (526 Dart files, 26 BLoC classes) – implemented contact, deal, activity, and project workflows with Google Maps location picker, Firebase analytics / crashlytics / remote config, multi-language i18n, and document handling (PDF, Excel, PPT).
- Built **D'Arrigo Brothers (DAB)**, a logistics check-in app with Firebase email auth, OTP validation, and a Chopper-generated REST client over a clean-architecture layout.

Dhaka Boss Pvt. Limited

Mar 2020 – Dec 2020

Junior Mobile Application Developer

- Shipped the **Dhaka Boss Merchant Dashboard** – Flutter merchant-side app covering product CRUD with image uploads, multi-status order tracking, payment settlement history, and pie-chart sales analytics with offline-connectivity detection.
- Initiated the customer-facing Dhaka Boss Shopping App and contributed to the Ticketing App (online ticket purchase) before transitioning roles.

PiLabs Bd. Limited

Oct 2018 – Jun 2019

Research Collaborator (CSE Thesis Project)

- Built the Android + SQLite client of the **Smart Car Parking System** (slot booking, parking release, admin interface) for a CSE thesis research collaboration; integrated system published in **IEEE RAAICON 2019** with a reported 95% detection accuracy.

Selected Projects

- **TimeAtlas** (Rust + Tauri 2 + React + SQLite – open source, MIT) – Personal, local-only time-awareness desktop app: 1 Hz active-window tracker loop in Rust, idle detection, local SQLite with upsert-aggregated activity buckets, React + TypeScript timeline UI; cross-platform single binary, no cloud, no team telemetry.
- **Axum Multi-tenant Starter** (Rust + Axum 0.7 + sqlx + Postgres – open source, MIT) – Production-ready multi-tenant SaaS API starter: shared-database tenant discriminator pattern, Argon2 password hashing, JWT auth with custom AuthUser extractor scoping every query, structured error mapping, JSON tracing, automatic sqlx migrate on startup.
- **Flutter ↔ Rust gRPC Demo** (Flutter + Rust + tonic + protobuf – open source, MIT) – Reference implementation for Flutter clients calling a Rust backend over gRPC: a single .proto source of truth, tonic server demonstrating unary + server-streaming + bidi-streaming RPCs (broadcast channel for live updates), Flutter client with strongly-typed contracts; protocols vendored.
- **Happy Restaurant Suite** (Flutter + Rust/Axum + PostgreSQL) – Multi-tenant restaurant SaaS: Rust backend (22 route modules, WebSocket kitchen-order push, recipe-driven inventory auto-deduction, multi-branch P&L) paired with a 34-screen Flutter admin app (KOT, procurement, staff scheduling) and a 15-screen Flutter customer app (QR table ordering, real-time delivery tracking, loyalty).
- **Inventory + CRM + POS SaaS** (Flutter + Rust/Axum + PostgreSQL) – Multi-tenant SaaS platform: Rust backend with 54 production REST endpoints, Docker-packaged, with subscription jobs, rate limiting, and audit logging; 144-file Flutter client covering multi-location warehouse, batch / lot tracking, supplier performance metrics, and financial analytics.
- **Qurbani & CWA** (Next.js + Express + Prisma + PostgreSQL – schema & architecture lead) – Designed PostgreSQL schemas (111 Prisma models, 96 migrations) and architectural blueprints for two web platforms: **Qurbani** (sacrificial-animal marketplace with Stripe + SSL Commerz, BullMQ jobs, Cloudinary uploads) and **CWA** (community-association management with multi-tenancy, Arabic/English + Hijri calendar, donations, member lifecycle, grave management, QR-coded ID cards).
- **Personal Manager + Rust Backend** (Flutter + Rust – open source: [client](#), [backend](#)) – Personal finance and task-management suite: Flutter mobile/desktop client paired with a Rust backend; SQLite storage, Excel/PDF export, offline-first design, reusable bulk-SMS Flutter module.

Education

B.Sc. in Computer Science & Engineering, United International University, Dhaka

Apr 2015 – Oct 2019

Higher Secondary Certificate, Govt. Shaheed Bulbul College, Pabna

Jan 2013 – Mar 2014

Secondary School Certificate, Pabna Zilla School, Pabna

Jan 2011 – Mar 2012

Publications

Smart Car Parking Management System, IEEE RAAICON

2019